

Chapter 4

RNN, LSTM, GAN

a brief introduction...

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Objective

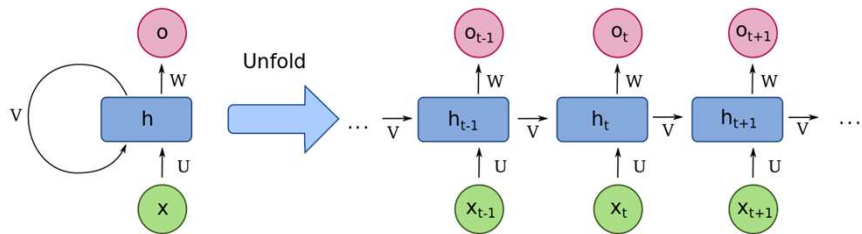
- Introduce the basic ideas of
 - Recurrent Neural Networks (RNN)
 - Long Short-Term Memory (LSTM) networks
 - Generative Adversarial Networks (GAN)
- No details will be addressed
 - Discover and learn: part of your project

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RNN



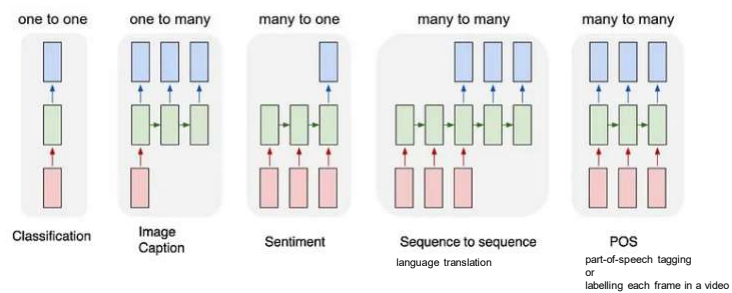
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<https://commons.wikimedia.org/w/index.php?curid=60109157>

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RNN Types



<https://medium.com/analytics-vidhya/in-depth-tutorial-of-recurrent-neural-network-rnn-and-long-short-term-memory-lstm-networks-3a782712a09f>

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RNN Problems

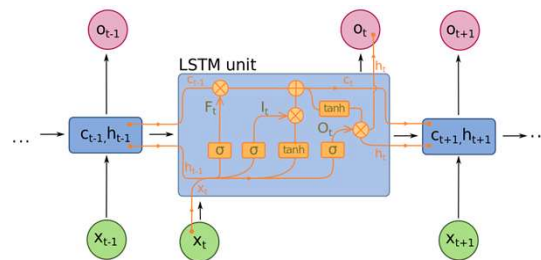
- Ineffective in learning long-range dependencies in sequences
 - Vanishing gradient
 - Exploding gradient
- LSTM: improved RNN

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LSTM



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Forget Gate: (F)

Determines what information from the cell state should be discarded or kept.

Input Gate: (I)

Controls the updates to the cell state and creates a candidate update that could be added to the cell state.

Output Gate: (O)

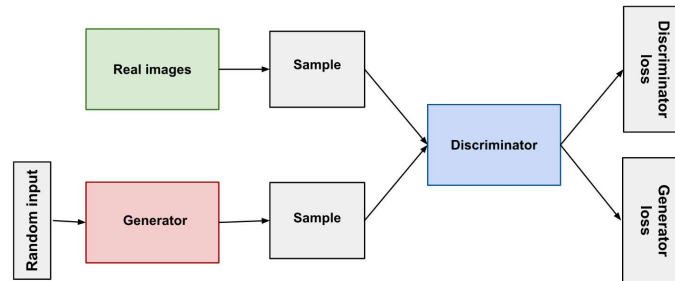
Controls the information that will be output from the cell state and produces the hidden state that will be passed to the next LSTM cell or used as the output.

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GAN



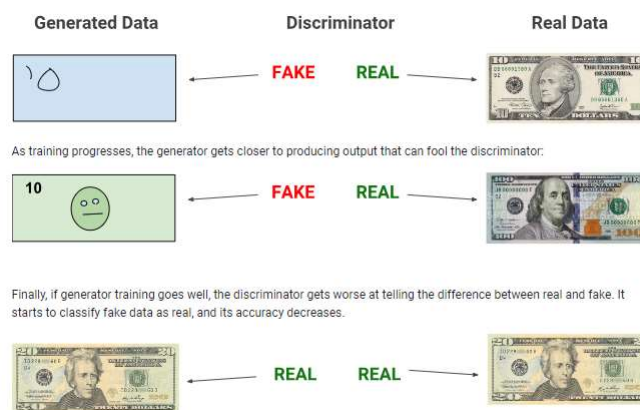
https://developers.google.com/machine-learning/gan/gan_structure

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GAN Training



https://developers.google.com/machine-learning/gan/gan_structure

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